

Chapter 17

AI in Environmental Social Work and Climate Change



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1 Introduction

The world is currently confronting an unprecedented convergence of social and environmental crises, including climate change, pandemics, rising global inequality, and geopolitical instability (International Federation of Social Workers [IFSW], 2022). These interconnected challenges are not only transforming human and ecological systems but are also worsening vulnerabilities among already marginalized populations. The climate crisis is intensifying existing social and economic inequalities and causing increased emotional, ecological, and financial distress across communities worldwide (IFSW, 2022; UN Security Council, 2021). The root causes of many of these crises are deeply rooted in historical and systemic structures, such as colonial exploitation and neoliberal economic policies (Billiot et al., 2019; Rao et al., 2022; Martinez & Irfan, 2021). Climate change poses a serious threat to both ecosystems and human well-being. It acts as a “crisis multiplier,” heightening conflicts over limited resources, increasing displacement, and driving humanitarian emergencies (UN Security Council, 2021).

Social workers are increasingly working with vulnerable populations and new technology tools, including artificial intelligence (AI), for their professional practice. However, social work interventions in the environment and in technology are limited, often focusing on disaster relief, rehabilitation, and advocacy during emergencies. Traditional social work has been criticized for focusing primarily on the

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381

social dimensions of a person's environment while neglecting the natural environment, which is essential to holistic well-being. The social work critique argues that the neoliberal economic model equates well-being with economic growth, but it has contributed to deepening injustice and neglecting holistic well-being (Boetto, 2017; Peeters, 2012; Powers et al., 2021). On the other hand, social work is far behind in adopting emerging technologies and in contributing to ethical innovations in AI for social justice. When it comes to AI and environmental social work, we are still in the ideation phase. Considering the limited literature and evidence-based practice, in this chapter, we try to provide a comprehensive knowledge and understanding of eco-social work, potential avenues for AI in environmental social work in response to climate change, highlighting some of the AI innovations for climate adaptation, the role of social work at the micro- and macro-levels, ethical considerations, and the way forward for environmental justice.

1.1 Eco-Social Work

Eco-Social Work, a form of environmental social work, is an emerging theoretical and practical framework that combines environmental sustainability with social justice, highlighting the interconnectedness of ecological and social systems (Boetto, 2017; Fato, 2024). This framework challenges traditional social work boundaries by recognizing that environmental issues directly impact human well-being, especially among marginalized populations (Gartshore, 2022; Matties et al., 2001; Närhi & Matthies 2016a, 2016b). The roots of Eco-Social Work stem from the Ecological Systems Theory (Bronfenbrenner, 1977), which views individuals as part of dynamic systems influenced by their environment. This perspective helps social workers understand the broader ecological contexts affecting communities and supports the creation of sustainable interventions (Coates & Gray, 2012; Peeters, 2012). Boetto (2017) outlines three levels: personal being, professional thinking, and practical doing. The individual level involves recognizing one's identity as connected to nature, influencing lifestyle choices. The thinking level calls for a transformation in professional values and ethics, including appreciation for traditional Indigenous knowledge and a commitment to global environmental justice (Billiot et al., 2019). The doing level involves implementing practice strategies that incorporate ecological considerations into daily social work interventions (Gartshore, 2022). Thysell and Cuadra (2023) further describe five types of eco-social work, ranging from a holistic approach, where human and ecological interests are inseparable, to various views that favor one dimension over the other.

Eco-Social Work takes a holistic view, recognizing that individual well-being is deeply influenced by environmental, social, cultural, and economic factors (Matties et al., 2001; Närhi & Matthies 2016a, 2016b; Powers et al., 2021). The Environmental Justice Framework (Lee, 1992; Van Horne et al., 2023) enhances Eco-Social Work by tackling the unequal distribution of environmental harms. It emphasizes how vulnerable groups often suffer the most from pollution and ecological damage.

Practitioners use this framework to promote equity and amplify the voices of those most affected (Schmitz et al., 2012). These interconnected factors influence access to vital resources, educational and job opportunities, social support networks, and the state of one's physical environment. Instead of focusing solely on individual challenges, this approach allows social workers to address both immediate issues and their systemic roots (Ungar, 2002). It also enables practitioners to challenge and change broader systems of inequality and environmental harm that disproportionately impact marginalized communities (Belchior Rocha, 2018). Additionally, this approach encourages society to use natural resources wisely, cut carbon footprints, and pursue overall well-being for everyone (Wang & Altanbulag, 2022).

This comprehensive lens supports more effective and sustainable interventions that are sensitive to the complex realities of people's lives. Intersectionality is fundamental to Eco-Social Work, as it recognizes the overlapping systems of oppression, such as race, caste, gender, sexuality, abilities, legal status, and class, that influence environmental vulnerabilities. By applying an intersectional perspective, Eco-Social Work seeks to create more inclusive and equitable pathways toward sustainable development (Norton, 2012; Shackelford et al., 2024). Although Eco-Social Work is still in its early stages, now is the right time to introduce AI and emerging technologies into the intervention process, as they can multiply their impact on ecosystems, communities, and vulnerable populations worldwide. The following sections will explore real-world AI innovation examples, potential opportunities, challenges, and ethical considerations for the effective integration of AI in Eco-Social Work.

2 Integrating AI in Eco-Social Work

Environmental social work, grounded in its enduring "person-in-environment" perspective, has traditionally focused on sociopolitical and economic systems. Yet in the past decade and a half, scholars have increasingly recognized the equally urgent importance of the natural environment in shaping human well-being (Mason et al., 2017; Rao et al., 2022; Shackelford et al., 2024). For frontline communities—especially those living in flood-prone coastal areas, wildfire corridors, or drought-affected agricultural regions—environmental degradation is not a distant abstraction. Environmental degradation, a daily, material concern, is interwoven with housing insecurity, food access, public health, and displacement (Billiot et al., 2019; Camps-Valls et al., 2025; Kumar et al., 2019; Yadama, 2013). In recent years, AI has increasingly been integrated into environmental efforts, creating new opportunities for disaster preparedness, resource management, and climate adaptation. AI-driven systems can analyze large volumes of environmental data, predict extreme weather events, optimize resource allocation, and support decision-making. AI's role in climate action is especially relevant to social work, as practitioners work to ensure equitable access to technology-driven solutions while protecting the rights of vulnerable populations (Ren & Wierman, 2024). Incorporating AI into Eco-Social

Work offers significant potential to improve both micro- and macro-level interventions focused on environmental sustainability and social justice. At the micro-level, AI can help social workers deliver personalized, data-driven interventions that account for ecological and social factors that affect well-being. For example, AI tools can analyze patterns in environmental and social data—such as housing quality, heat islands, pollution exposure, or access to green spaces—to identify at-risk individuals and guide early intervention efforts (Naslund et al., 2020). These tools can also enable remote counseling, digital case management, and predictive modeling to better understand how ecological issues, such as climate change, disproportionately impact marginalized groups (Coates & Gray, 2012). At the macro-level, AI can support policymakers and organizations in identifying and tackling systemic environmental injustices. AI applications in environmental monitoring, resource allocation, and climate risk assessment can inform data-driven policies that prioritize equity and sustainability (Närhi & Matthies 2016a, 2016b). For instance, AI-powered geographic information systems can map areas experiencing high ecological degradation and socioeconomic disadvantage, facilitating targeted interventions and fair resource distribution (Peeters, 2012). Macro-level social workers can partner with technologists to ensure AI systems' design and development reflect diverse perspectives and needs, especially those of historically marginalized communities (Mathiyazhagan, 2021).

Additionally, AI can bolster advocacy efforts by creating compelling data visualizations and predictive scenarios that highlight the urgency of eco-social issues to stakeholders and the public (Schmitz et al., 2012). However, practitioners must remain vigilant in ensuring responsible AI use, including safeguarding privacy and avoiding algorithmic bias that could deepen marginalization of vulnerable groups (Jatobá et al., 2021). Incorporating AI into Eco-Social Work requires a deliberate commitment to ethical practices, inclusivity, and sustainability. AI tools must be designed, developed, and implemented with input from affected communities to ensure cultural relevance and reduce technological exclusion.

3 AI and Climate Change

AI-powered climate models analyze vast datasets from satellites, sensors, and meteorological sources to identify patterns and predict long-term climate trends (Olawade et al., 2024). One notable example is the International Methane Emissions Observatory, which leverages AI to monitor methane emissions, a potent greenhouse gas contributing to global warming (Rolnick et al., 2022); United Nations Environment Programme [UNEP], 2025). Machine learning algorithms also enhance forecasting accuracy for hurricanes, wildfires, and floods, allowing for more timely interventions (Wang et al., 2023). The European Centre for Medium-Range Weather Forecasts (ECMWF) (n.d.) has developed AI-driven climate models that improve predictions of extreme weather events, helping policymakers and emergency responders prepare more effectively (Zewe, 2025).

AI tools are also increasingly embedded in disaster management. For example, Marine Sand Watch (UNEP, [n.d.](#); UNEP, [2024](#)) uses AI to track sand extraction from marine environments, helping to prevent ecosystem destruction. AI-powered satellite imaging can detect rising water levels and predict flooding patterns, improving response time and mitigating damage (Wang et al., 2023). Similarly, AI models analyze temperature, wind patterns, and vegetation dryness to anticipate wildfires, improving wildfire prediction efforts (Ren & Wierman, [2024](#)). In hurricane-prone regions, AI-driven simulations enhance hurricane path prediction and emergency planning, aiding disaster response teams (ECMWF, 2024). For eco-social workers engaged in crisis intervention, AI can provide real-time data on affected populations, identify the most vulnerable areas, and help streamline aid distribution (Dominelli, [2018](#)).

3.1 Resource Management

AI's contributions to resource management and climate adaptation are extensive, offering innovative solutions to address environmental challenges. AI optimizes natural resource allocation by managing water distribution and detecting leaks to conserve resources (Kamyab et al., [2023](#)). Traditional methods of water management often rely on manual inspections and historical data, which can be inefficient and prone to errors. AI-driven systems, however, can analyze real-time data from sensors and satellite imagery to identify leaks, predict water demand, and optimize distribution networks. Smart water grids, for instance, use machine learning algorithms to detect anomalies in water pressure and flow, allowing for rapid intervention and minimizing water loss.

These systems can also forecast water demand based on weather patterns, population density, and seasonal variations, ensuring efficient allocation and preventing shortages. Similarly, AI enhances energy efficiency in smart grids by balancing supply and demand, reducing waste, and improving sustainability (Zewe, [2025](#)). Traditional energy grids often struggle to adapt to dynamic supply and demand patterns, especially when integrating renewable energy sources. AI-powered smart grids can analyze real-time data from sensors and smart meters to predict energy needs, optimize energy distribution, and detect grid disturbances. These systems contribute to reliable energy access, which is crucial for community resilience.

3.2 Land Use Planning

AI supports land-use planning for sustainable urban development by analyzing geographic and demographic data to inform infrastructure projects (Ren & Wierman, [2024](#)). Urban planning often involves complex decisions with long-term environmental and social consequences. AI can aid urban planners by analyzing large

datasets on population density, traffic patterns, environmental conditions, and economic activity (Son et al., 2023). For example, AI can identify optimal locations for new infrastructure projects, such as public transportation, parks, and affordable housing, minimizing environmental impact and maximizing social benefits (Cina et al., 2025).

Moreover, AI can assess the vulnerability of different areas to climate change impacts, such as flooding, heatwaves, and sea-level rise, and recommend appropriate adaptation measures (Camps-Valls et al., 2025; Yadama, 2013). For environmental social workers, these tools can support efforts to advocate for climate-resilient infrastructure in underserved neighborhoods, ensure inclusive community engagement in planning processes, and safeguard vulnerable populations from displacement (Billiot et al., 2019).

3.3 Localized Climate Adaptation Strategies and Predictive Analytics

For marginalized communities disproportionately affected by environmental degradation, AI can provide tailored solutions, such as localized climate adaptation strategies (Rolnick et al., 2022). Climate change impacts are often felt most acutely by marginalized communities, who may lack the resources and infrastructure to adapt to changing conditions (Mason et al., 2017; Rao et al., 2022). AI can help these communities by providing localized climate adaptation strategies tailored to their specific needs and circumstances.

Predictive analytics powered by AI are enabling a new scale and scope of environmental modeling for climate adaptation. In water-scarce regions, for example, AI applications in climate-resilient water management have demonstrated significant potential to improve adaptation outcomes. Abdulameer et al. (2025) found that long short-term memory (LSTM) and hybrid physics-informed neural networks achieved higher accuracy in hydrological forecasting (Nash–Sutcliffe efficiency >0.90), enabling early warnings for droughts and floods. In agricultural settings, Internet-of-Things-integrated AI systems reduced water waste by 20%–40% and optimization models improved the distribution efficiency by up to 30%. Yet the study also highlighted significant challenges: 70% of AI tools remain concentrated in data-rich, temperate zones, leaving arid and low-income communities largely underserved. These disparities underscore the need for decentralized data ecosystems and policy alignment to ensure marginalized communities benefit from AI interventions.

While traditional farming practices often rely on generic approaches that can be inefficient and environmentally damaging, precision agriculture systems use AI to analyze data on soil conditions, weather patterns, and crop health to optimize irrigation, fertilization, and pest control (Wang et al., 2023). For example, Hultgren et al. (2025) projected climate change impacts on agricultural yields across 24,378 global

regions using 33 climate models and nearly 2000 Monte Carlo simulations per crop. These projections account for adaptation, income growth, and CO₂ fertilization, while providing granular insights invaluable to environmental social workers addressing food insecurity in climate-stressed communities. These systems can also provide farmers with real-time recommendations on when to plant, harvest, and rotate crops, maximizing yields and minimizing environmental impact (Pandian, 2025). For environmental social workers, such insights are crucial for advocating equitable resource governance while co-designing interventions that build trust in technological systems.

4 AI Intervention for Climate Adoption

The AI-driven eco-social work models enhance marginalized community capacity to support disaster preparedness, advocate for policy changes, and promote sustainable solutions. AI-driven disaster response systems can assist social workers in tracking displaced individuals, coordinating emergency housing, and ensuring equitable resource distribution (Albahri et al., 2024; Haykal et al., 2025; Zewe, 2025). AI-driven wildfire detection systems help social workers support displaced communities by coordinating shelter, medical aid, and mental health services for affected individuals in California. AI-enhanced water management systems help social workers address drought crises by ensuring an equitable distribution of water resources and reducing the burden on impoverished communities. AI-driven deforestation monitoring enables social workers and Indigenous rights advocates to track environmental destruction, protect vulnerable communities, and push for policy interventions. AI-powered flood prediction models assist NGOs and social workers in implementing preemptive measures, reducing harm to vulnerable populations by facilitating early evacuations and resource allocation. The following real-world examples highlight AI-powered disaster preparedness and climate adaptation worldwide, exploring the possible role of eco-social workers.

4.1 Wildfire Detection Systems in California, USA

Many wildfire-prone regions like California face an escalating wildfire crisis due to prolonged droughts, rising temperatures, and land-use changes driven by climate change. In the last decade, the state experienced 15 of the 20 most destructive wildfires in its history, with over 11 million acres burned between 2017 and 2021 alone (California Department of Forestry and Fire Protection [CAL FIRE], 2025). These fires have displaced more than 150,000 people, destroyed thousands of homes, and disproportionately impacted rural, low-income, and Indigenous communities just in 2025 (Batko & McTarnaghan, 2025). Nonetheless, AI-driven wildfire detection systems are transforming how early warnings and resource deployments are handled.

Traditional wildfire monitoring, which relied heavily on human surveillance, satellite imagery, and static sensor networks, often faced delays in detection and had limited accuracy, especially in remote areas (Alkhatib, 2014; Di Giuseppe et al., 2025; Jen, 2024). However, AI-enabled wildfire detection systems analyze real-time data from ground sensors, drones, and satellite imagery to identify ignition points and smoke plumes within minutes—far faster than human responders. Programs like ALERTCalifornia, Google's FireSat, IBM's Geospatial Analytics, and University of Southern California's conditional Wasserstein Generative Adversarial Network (cWGAN) model use machine learning to detect anomalies and predict fire spread based on variables such as wind speed, humidity, and vegetation density (ALERTCalifornia, 2025; Brodsky, 2025; Di Giuseppe et al., 2025; Google Research, n.d.-a, n.d.-b; IBM Environmental Intelligence Suite, 2025; Jen, 2024; Raffio, 2024).

One of the most impactful implementations has been the ALERTCalifornia platform at the University of California, San Diego, recognized as one of TIME Magazine's best inventions of 2023 (Dixit, 2023). This system uses over 1100 high-definition cameras paired with AI algorithms to detect smoke plumes and analyze environmental risk factors across the state. Combined with satellite data and community-based response apps, ALERTCalifornia has helped reduce emergency response times and improve coordination among emergency services, social workers, and local governments. The system works by flagging potential fire outbreaks, which are then rapidly reviewed by trained personnel, reducing false positives and cutting detection time to under 60 s in many cases (ALERTCalifornia, 2025).

AI-driven wildfire detection systems also help social workers support displaced communities by coordinating shelter, medical aid, and mental health services for affected individuals (Ren & Wierman, 2024). By forecasting fire spread and high-risk zones, these systems enable preemptive evacuations and resource distribution, minimizing chaos during crises. They allow social workers and local responders to use predictive mapping to direct at-risk residents, such as the older and undocumented individuals, to safety before evacuation routes close (Bell, 2021). AI-enabled alerts also help agencies anticipate housing, food, and counseling needs in advance, rather than responding reactively once a fire has already displaced people. Recent advancements in deep learning have significantly improved the accuracy and speed of wildfire detection using satellite imagery (Matias, 2023).

Another notable development was the creation of the California Wildfire GeoImaging Dataset (CWGID), a large-scale, annotated dataset comprising more than 100,000 pairs of high-resolution, temporally matched satellite images (Martin et al., 2024a). This system captures the same geographic area at different times for each image pair, enabling AI models to learn how wildfires visually manifest and evolve across landscapes. The dataset also includes pre-fire, active fire, and post-fire imagery, providing a robust foundation for supervised learning (Martin et al., 2024b). Using this dataset, researchers implemented EfficientNet-B0, a convolutional neural network architecture known for its optimal trade-off between accuracy and computational efficiency (Klingler, 2024). The model achieved over 92% accuracy in wildfire detection tasks, outperforming traditional image classifiers and

demonstrating robust generalization across diverse terrain and environmental conditions (Martin et al., 2024a). The success of such models lies in their ability to detect subtle changes in thermal signatures, smoke trails, and vegetation reflectance that may not be visible to human observers or older detection systems (Giannakidou et al., 2024).

Beyond emergency response, AI technologies are also supporting longer-term social work efforts. By identifying fire-prone zones and socioeconomically vulnerable populations, AI can inform resilience-building programs—such as fire-resistant infrastructure upgrades, insurance subsidies, and trauma-informed mental health services tailored to disaster survivors (Ren & Wierman, 2024). Research shows that anticipatory action, when supported by innovations such as AI-driven forecasting, can significantly reduce disaster-related losses and strengthen community preparedness by improving the effectiveness of early warning systems (de Pérez et al., 2022; United Nations Framework Convention on Climate Change [UNFCCC], 2024). Thereby, California's integration of AI into wildfire detection offers a compelling model for combining environmental science and eco-social work service delivery.

4.2 Flood Prediction Models in Bangladesh

Traditional flood warning systems often rely on historical data and manual monitoring, which can be slow and inaccurate (Nearing et al., 2024). AI-powered flood prediction models can analyze real-time data from weather stations, river gauges, and satellite imagery to predict flood events with greater accuracy and speed (Liu et al., 2024). These models, such as FloodSENS, MIT's Earth Intelligence Engine, and Google's Flood Hub, can also identify the most vulnerable areas to flooding, allowing social workers to target interventions and prioritize evacuations (The European Space Agency [ESA], 2020; Chu, 2024; Matias, 2021).

In 2018, Google launched the Flood Forecasting Initiative, an AI-driven system designed to combine hydrologic models (predicting water flow) with inundation models (predicting flood spread), blending conventional physics-based approaches with machine learning trained on satellite and river data to improve the accuracy and reach of flood early warning systems (Matias, 2021; Nevo et al., 2022). Nevo et al. (2022) explain how Google's operational flood warning system uses LSTM networks and manifold models to predict both river stage and inundation depth with high precision. During the 2021 monsoon season, this system covered 470,000 km² in India and Bangladesh, issuing over 100 million alerts to over 350 million people. Crucially, these alerts were delivered not only to government authorities but also directly to residents via mobile platforms, including community-based WhatsApp groups and SMS systems accessible in low-bandwidth settings. These timely alerts, often provided up to five days in advance, helped residents evacuate safely, protect livestock, and secure property (Chowdhury, 2021; Liu et al., 2024; Nevo et al., 2022).

Moreover, Google expanded this initiative by launching Flood Hub, a platform designed to provide real-time flood forecasts and early warnings to over 100

countries, and reaching more than 700 million people worldwide (Google Research, 2025; Matias, 2024; Nearing et al., 2024). Studies and field reports have shown that the system has increased trust in flood warnings, improved response coordination among local agencies, and reduced flood-related fatalities by 43% (Chowdhury, 2021; Nearing et al., 2024). Notably, in the Jamuna River basin of Bangladesh, organizations like GiveDirectly have used Flood Hub's forecasts to make anticipatory cash transfers to vulnerable communities before the July flood season, allowing families to safeguard their homes and livelihoods (ESG News, 2025). Flood Hub exemplifies how AI can bolster climate resilience, especially in regions lacking extensive hydrological infrastructure. By providing precise and timely flood forecasts via open access, Flood Hub illustrates how predictive analytics can strengthen disaster response at scale, empowering communities to mitigate climate-related disasters and supporting researchers and policymakers in developing more effective disaster preparedness and response strategies (Google Research, 2025).

4.3 Rainforest and Deforestation Monitoring in the Amazon

Deforestation in the Amazon rainforest is a major environmental and social problem, threatening biodiversity, climate stability, and the livelihoods and autonomy of Indigenous communities. Forests cover 31% of global land area, with 497 million hectares or 12.2% of this forest area found in Brazil (Abdenur, 2020; Food and Agriculture Organization of the United Nations [FAO], 2020). However, deforestation accounts for 12%–20% of global greenhouse gas emissions (FAO, 2020; Gerber & Foley, 2025). Driven primarily by illegal activities including induced forest fires, illegal mining and logging, and land invasions, the commodification and degradation of forests are further expedited by legal agriculture and livestock production industries (FAO, 2020, 2022; Gerber & Foley, 2025).

As local communities and Indigenous Peoples either own or manage nearly half, 4.35 billion hectares, of the world's forest and farm landscapes, yet contribute nothing to deforestation, working with Indigenous rights advocates and social workers to track environmental destruction, protect vulnerable communities, and push for policy interventions is crucial for deforestation monitoring (FAO, 2022; Rolnick et al., 2022). Traditional forest monitoring methods often relied on ground surveys, delayed satellite data, or manual reporting, which were insufficient given the scale and speed of forest loss (Global Conservation, 2024; Global Forest Watch, n.d.; Ennes & Chaves, 2021). However, these ground-based methods also provide critical insights: communities that depend on forests for food, fiber, and income often detect ecological changes long before satellite imagery does. For example, in dry tropical forests of the Eastern Ghats, researchers partnered with local communities to assess the impact of harvesting on endemic tree species. Their findings revealed species-specific vulnerabilities and the demographic consequences of non-timber harvesting—information essential to conservation and predictive modeling (Neeraja et al., 2022). Community-led forest health monitoring provides granular, context-specific

data, including threats such as fire, soil erosion, and grazing, which may not be visible through remote sensing alone. These insights are also relevant to IUCN red-list assessments and sustainable harvesting guidelines.

In contrast, AI systems analyze satellite imagery, audio inputs, and predictive models to detect illegal logging early, alert indigenous groups, and guide policy response (Rolnick et al., 2022). Collaborative platforms like PrevisIA, developed by the Brazilian NGO Imazon and Microsoft, analyze high-resolution satellite imagery and land-use patterns to identify unauthorized roads and other indicators of deforestation risk in real time (Imazon, 2022; PrevisIA, n.d.; Smith, 2023). Similarly, Rainforest Connection (RFCx) uses real-time bioacoustic monitoring to combat illegal deforestation in the Amazon and other biodiverse regions. The non-profit utilizes recycled, solar-powered recording devices known as “Guardians,” which are installed high in the forest canopy of protected territories to continuously capture environmental sounds, including chainsaws, gunshots, and vehicle engines (Rainforest Connection, 2025). RFCx’s technology transmits immediate alerts to local first responders, NGOs, and Indigenous forest rangers via mobile networks (Goldapple, 2021; Rainforest Connection, 2025). These sensors also generate a forensic record of illegal activity that can be used in court (Goldapple, 2021). In Brazil, indigenous groups such as the Temb  have adopted the system to strengthen their surveillance and document invasions for use in legal and advocacy efforts (Abdenur, 2020).

Moreover, the platform promotes a participatory and rights-based model by keeping data control within the community and supporting self-determined environmental governance (Abdenur, 2020; Prestes et al., 2024). The integration of AI with community-based knowledge and field action has not only enhanced the precision of forest monitoring but also empowered indigenous populations to protect their land more effectively (Abdenur, 2020). Open-access tools like Global Forest Watch further amplify these efforts by providing transparent, actionable data to researchers, policymakers, and activists worldwide (Abdenur, 2020; Global Forest Watch, n.d.). These examples exemplify how AI, when ethically and inclusively deployed, can empower marginalized groups and support environmental justice.

4.4 Drought Monitoring in Sub-Saharan Africa

Sub-Saharan Africa faces increasingly frequent and severe droughts due to anthropogenic climate change, affecting the livelihoods of over 61 million people (International Centre of Expertise in Montreal on Artificial Intelligence, 2024; United Nations Office for the Coordination of Humanitarian Affairs [UN OCHA] & Regional Interagency Standing Committee [RIASCO], 2024). An estimated \$71 billion has been lost over the last 50 years, disproportionately affecting smallholder farmers, pastoralists, and other communities that rely on subsistence farming, with about 23% of sub-Saharan Africa’s GDP coming from agriculture (Goedde et al.,

2019; Okoth, 2024). Traditional drought monitoring methods—such as rain gauge data, farmer surveys, or manual soil assessments—are often reactive, fragmented, and lack regional coordination (Le Coz & van de Giesen, 2020). These limitations hinder timely interventions, leaving vulnerable populations without adequate early warning, crop planning tools, or water resource management.

The human toll of these droughts is profound. In 2024, Southern Africa experienced its worst drought in decades, exacerbated by El Niño, leaving over 27 million people facing acute food insecurity and 21 million children suffering from severe malnutrition (Imray, 2024; Sinyangwe & Savage, 2024). Countries such as Zambia, Malawi, and Zimbabwe declared national disasters as crops failed and water sources dwindled (Khorsandi, 2024). Social workers play a critical role in ensuring these populations receive timely, culturally sensitive support, yet fragmented data and delayed crisis recognition often hinder their efforts. However, AI-informed early warning systems allow practitioners to shift from reactive to anticipatory responses, enabling the mobilization of cash transfers, psychosocial support, nutritional aid, and water provision before conditions deteriorate (Kühnle, 2024).

AI-driven models also integrate satellite imagery, weather forecasts, and soil moisture data to provide early warnings and inform resource allocation (International Centre of Expertise in Montreal on Artificial Intelligence, 2024; Sheffield et al., 2014). For instance, the African Flood & Drought Monitor utilizes machine learning algorithms to deliver real-time drought assessments and up to 6 months of forecasts, aiding farmers, social workers, and policymakers in timely decision-making processes (Sheffield et al., 2014; United Nations Convention to Combat Desertification [UNCCD], n.d.). Additionally, the Food and Agriculture Organization's Water Productivity Open-access Portal (WaPOR) uses remote sensing and AI analytics to provide near real-time data on evapotranspiration, biomass production, and water use efficiency across Africa and the Near East (FAO, 2024). These insights help governments and humanitarian agencies manage irrigation systems, prioritize high-risk areas, and optimize land productivity during droughts.

A notable case study is the implementation of AI-based drought prediction models in Kenya. Researchers developed an LSTM neural network to forecast vegetation health using satellite-derived indices. This model achieved an accuracy of 78.3% in predicting drought conditions, enabling timely interventions for pastoral communities dependent on livestock and agriculture (Barrett et al., 2019). The benefits of anticipatory action using AI models are substantial. Improved early warning systems have contributed to a threefold decrease in deaths from weather, climate, and water-related hazards over the past 50 years, despite an increase in the frequency of such events (World Meteorological Organization [WMO], 2024). Taken together, these AI-powered tools help bridge the gap between early warning and early action, offering data-driven support for drought resilience efforts in one of the world's most climate-vulnerable regions. These examples demonstrate AI's potential to enhance social work efforts in climate-related interventions. However, AI

solutions must be designed inclusively to prevent exacerbating inequalities and ensure equitable access to resources.

4.5 Ethical Considerations for Environmental Justice

AI has the potential to advance environmental justice by tracking pollution in low-income areas and holding corporations accountable for environmental harm (Ren & Wierman, 2024). The rapid development of AI raises serious ethical concerns regarding data ownership, privacy, and the risk of surveillance disproportionately targeting marginalized communities (Zewe, 2025). AI-driven environmental monitoring tools must be designed with transparency and accountability to prevent the reinforcement of existing environmental injustices (Li et al., 2025). Data ownership and privacy are major concerns, as AI systems often collect and analyze large amounts of personal data. AI-driven environmental monitoring tools could be used to monitor the activities of activists and community organizers, chilling free speech and undermining environmental justice efforts. To prevent these problems, AI-driven environmental monitoring tools must be co-designed with impacted communities with complete transparency and collective accountability. This includes ensuring that data are collected and used responsibly, and that communities have meaningful opportunities to participate in how AI systems are developed and deployed.

4.5.1 Technology Access Gap

Ensuring equitable access to AI-driven resources remains a challenge that must be addressed through inclusive policymaking and equitable AI governance (Dominelli, 2018). Marginalized communities often lack the resources and infrastructure to access and benefit from AI-driven solutions. Addressing this challenge requires capacity building, infrastructure development, access to advanced devices, and inclusive policymaking that prioritizes the needs of marginalized communities and ensures equitable resource allocations to enable access to AI technologies. Equitable AI governance is also essential to prevent AI from exacerbating existing inequalities. This includes ensuring that AI systems are designed, developed, and deployed in a transparent and accountable manner, and that the voices of marginalized communities are heard in decision-making processes.

4.5.2 Bias in AI Models

AI-driven environmental monitoring may reflect biases in training data, leading to inaccurate or unjust outcomes (Ren & Wierman, 2024). AI models rely on extensive datasets, but biases in data collection can lead to erroneous predictions that disproportionately impact marginalized communities (Li et al., 2025). For example,

suppose an AI-driven environmental monitoring system is trained on data that underrepresents pollution levels in low-income areas. In that case, it may underestimate the environmental risks these communities face.

4.5.3 The Digital Divide

Access to AI-driven climate solutions is uneven, often favoring wealthier nations and communities while leaving others vulnerable (Zewe, 2025). The digital divide refers to the gap between those who have access to digital technologies, such as computers and the Internet, and those who do not. This divide can exacerbate existing inequalities, as marginalized communities may lack the resources and infrastructure to access and benefit from AI-driven climate solutions.

4.5.4 Environmental Impact of AI

The environmental impact of AI itself cannot be overlooked. AI systems rely on critical minerals, contributing to mining-related environmental degradation (Corrigan & Ikonnikova, 2024). The computational demands of AI contribute to environmental degradation, as training large-scale models requires substantial energy and resources (Olawade et al., 2024). The production and disposal of AI hardware generate electronic waste containing toxic substances like lead and mercury (Corrigan & Ikonnikova, 2024; Iqbal, 2024). AI consumes significant electricity, with data centers requiring extensive energy resources. Moreover, CO₂ emissions and water usage in AI operations are another primary concern. For instance, Google's U.S. data centers alone consumed an estimated 12.7 billion liters of fresh water in 2021 to cool servers (Kirkpatrick, 2023; Li et al., 2025). It is essential to consider energy-efficient AI models to reduce the environmental impact of AI technologies.

5 Role of Social Work

Social workers play a critical role in climate change preparedness and in supporting vulnerable communities in adopting climate change measures. Social workers, deeply committed to principles of justice, equity, and human rights, play a vital role in addressing the complex challenges of climate change through eco-social work practice (The Social Work Graduate, 2023; Timariu, 2022; Ungar, 2002). As climate-related disasters grow in frequency and intensity, integrating AI offers transformative potential to enhance social work's impact on disaster preparedness, adaptation, and policy development. Through ethical application, community engagement, and rights-based advocacy, social workers can ensure that AI is harnessed not merely as a technological tool but as a catalyst for equity-driven, sustainable transformation. In this section, we envision the role of social work in three

levels: (1) the role of social work in AI system development, (2) the use of AI in eco-social work practice, and (3) addressing the digital divide for digital inclusion.

5.1 Role of Social Workers in AI Development

As a practice-centric profession grounded in transdisciplinary theories, the adoption of indigenous knowledge, and ongoing community engagement, social workers play a critical role in the AI development cycle for climate change preparedness tools by incorporating human rights and social justice values to ensure AI models and tools are inclusive, equitable, and socio-culturally responsive. Social workers also act as advocates for vulnerable populations and actively incorporate indigenous knowledge and community perspectives to align AI systems with community needs and ethical standards throughout the development process (Harms, 2007). Table 17.1

Table 17.1 Role of social workers in the AI development process

AI development cycle	Role of social workers
Planning	Help identifying high risk and vulnerable population and meaningfully engage them in the planning process to make sure that AI tools meet the community needs and reduce disproportionate impact to the vulnerable populations. Support in required capacity building of communities. Equitable compensation model for community engagement.
Design	Adapt design justice principles for meaningful engagement of communities and support co-design process with communities, designers, and technologies to ensure design frameworks meets the community needs.
Data collection	Support to ensure that data are gathered ethically and reflects the lived realities of diverse communities impacted by climate related disasters. Ensure under/over-representations of certain social groups.
Data analysis	Support developers to look for people behind the data and help them understand local socioeconomic cultural contexts, sensitivities, and barriers in accessing resources and technologies.
Model development	Collaborate with AI developers to identify possible bias, harms, and unintended consequences of AI predictive models.
Validation	Support to validate the AI models and tools with communities to ensure diversity, inclusion, accessibility requirements and ensure that intended people can use it with context.
Implementation	Act as bridge between communities and AI developers. Support with education, capacity building and outreach to promote the responsible use of tools.
Evaluation	Assessing social and environmental impact, gathering community feedback, and recommending adjustments to ensure fairness, usability, and effectiveness.
Use of data	Ensure user data is not used beyond indented purpose and equitable compensation for the community data. Support to reduce carbon footprint, e-waste, energy conservation.

explains the role of social workers in each AI development cycle to ensure that climate change preparedness tools become more just, responsive, and impactful in building community resilience.

5.2 Use of AI in Eco-Social Work Practice

Social workers can leverage AI in multiple dimensions of climate change mitigation and adaptation. At the micro-level, AI-powered tools such as predictive analytics can help identify individuals and households most vulnerable to climate-related risks, enabling targeted interventions. For example, machine learning algorithms can assess socio-environmental data—such as health records, housing stability, and geographical exposure—to anticipate impacts of heatwaves, floods, or wildfires (Vinuesa et al., 2020). Social workers can then mobilize resources and develop personalized action plans to enhance individual and community resilience. On the mezzo level, social workers collaborate with community organizations to implement AI-driven early warning systems and decision-support tools. These systems can provide real-time risk assessments and guide evacuation strategies, resource allocation, and community education. Social workers play a critical role in translating complex data into accessible information and ensuring that marginalized populations—often excluded from technological benefits—receive timely and culturally appropriate support (Bulkeley et al., 2013). At the macro-level, social workers advocate for inclusive and ethical AI policy frameworks in disaster risk reduction and climate adaptation. They engage with governments and global organizations to ensure that AI technologies do not perpetuate existing social inequities. AI tools used in environmental planning and disaster management must be co-designed with communities, centering local knowledge and participatory governance (Eubanks, 2018). Social work's commitment to human rights and social justice ensures that technological innovation aligns with the broader goal of building a new eco-social world where no one is left behind. Moreover, integrating AI into eco-social work practice supports capacity-building and long-term sustainability. Training programs and interdisciplinary collaborations can equip social workers with digital literacy and technological fluency, enabling them to meaningfully contribute to AI development and deployment (IFSW, 2022). This integration reinforces the role of social workers as facilitators of social and environmental protection, advancing policy innovation to address the intersecting crises of climate change, inequality, and systemic disempowerment.

5.3 Addressing Digital Divide for AI-Supported Climate Adoption

Digital divide is a human rights issue (Sanders & Scanlon, 2021). Social workers' intervention is essential for AI solutions to reach and benefit all segments of society, particularly vulnerable and marginalized communities. Social workers actively work with vulnerable populations and marginalized communities. There has been a social voice calling for action to address the digital divide. To make AI in eco-social work practice a reality, it is vital to educate communities, and social workers are better positioned to help individuals understand and use AI tools designed for climate risk alerts, resource management, or early warning systems, ensuring that no group is left behind due to a lack of digital skills. When AI tools are introduced, social workers assist individuals and communities in navigating transitions, reducing resistance, and encouraging adoption to reduce climate-related risk and damage. Social workers advocate for equitable access to digital infrastructure and for fair resource allocation. Social workers collaborate with policymakers to prioritize Internet connectivity, device access, and inclusive digital services in underserved areas—prerequisites for communities to engage with AI-driven climate tools.

6 Conclusion

Environmental social work is increasingly grappling with the intensifying complexities of climate change, not only as an ecological crisis but also as a profoundly social one. As rising temperatures, violent storms, wildfires, and chronic water scarcity disproportionately impact vulnerable populations, practitioners are being called to address these cascading threats through both immediate and anticipatory frameworks. The integration of AI into eco-social work holds significant, transformative promise—not as a detached technological fix, but as a part of a hybrid approach that complements Indigenous and community-based knowledge systems. When combined, these knowledge forms can generate more comprehensive strategies for climate adaptation and resilience. For instance, AI-driven flood forecasting models or deforestation monitoring platforms are most effective when interpreted and operationalized alongside local observations, cultural practices, and social work's relational expertise. The focus, then, is not solely on demonstrating that AI can be beneficial but on identifying emerging examples where AI-enabled insights can be translated into concrete interventions at multiple levels: warning systems that help households evacuate or safeguard assets, adaptation strategies that amplify community resilience, and landscape-scale monitoring that supports policy and ecosystem-level interventions. Yet challenges remain. Gaps in the literature highlight the need for participatory AI approaches that actively involve community stakeholders in design and governance (Rolnick et al., 2022). Strengthening interdisciplinary collaborations between social workers, AI developers, and

environmental scientists is essential to align technological solutions with social and ecological goals (Dominelli, 2018; Mason et al., 2017). Moreover, policies must ensure that AI-driven environmental interventions are ethical, inclusive, and sustainable (Zewe, 2025). As AI continues to evolve, the credibility of this field will depend on moving beyond broad claims and instead demonstrating where and how AI, environmental work, and social work intersect in practice, particularly at the community and household level, where climate impacts are most acutely felt. By integrating Indigenous knowledge with predictive analytics and community-based social work, environmental interventions can become more accurate, just, and durable, offering pathways to resilience that ensure equitable climate solutions.

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